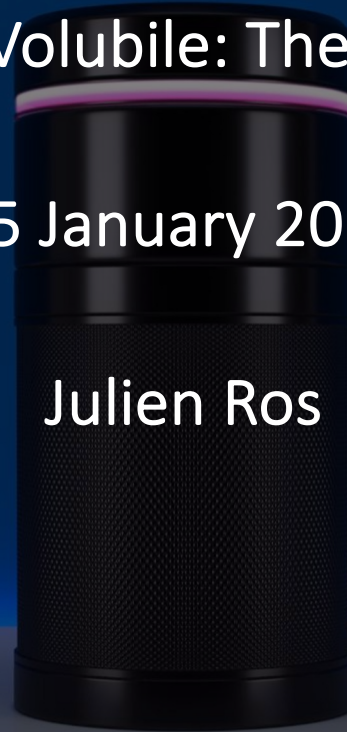


From Inception to Volubile: The Journey of Voice AI

15 January 2025

Julien Ros



Agenda

Volubile

- ✓ **Introduction**
- ✓ **The conversational loop**
- ✓ **Core components**
- ✓ **Handling complex conversations**
- ✓ **Other challenges**



Agenda



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Importance of Voice Technologies



Still a Strategic Channel

- ✓ 70% of complex customer interactions still happen over the phone.
- ✓ Preferred for urgent issues, sensitive matters, and high-stakes decisions.

Why It Matters

- ✓ Builds trust through real-time, human-like interaction.
- ✓ Critical for sectors like healthcare, finance, and emergency services.



<https://www.fortunebusinessinsights.com/industry-reports/contact-center-software-market-100840>

✦ Limitations of Traditional Solutions

IVR Systems (Interactive Voice Response)

- ✓ Rigid decision trees → poor flexibility.
- ✓ Users forced to follow predefined paths or press keys.
- ✓ Long setup times (weeks to implement).

Static Scripts

- ✓ Cannot handle unexpected questions or context changes.
- ✓ Lack of personalization → generic, robotic experience.

Impact on Customer Experience

- ✓ Frustration due to unnatural conversations.
- ✓ High abandonment rates during calls.
- ✓ Limited ability to resolve complex issues.

Operational Challenges

- ✓ Expensive to maintain and update.
- ✓ Scaling requires more human resources, not smarter technology.



<https://www.voicesage.com/blog/ivr-call-centre-problems/>

✦ Breakthrough with LLMs

Volubile[✦]



Good Conversationalists

If you've spent time in free-form dialog with ChatGPT or Claude, you know it feels **natural** and **broadly useful**.

- ✓ Natural conversation flow
- ✓ Human-like interactions



Excel at Structuring Data

They can turn **messy, unstructured input** into **clean, structured information**.

- ✓ Data extraction from conversations
- ✓ Organize key details from calls

Combine These Strengths

Natural conversation + Data extraction = New Voice AI systems

💫 3 Types of Company Shaping the Voice AI Industry

Volubile⁺

Foundation Model Companies

🧠 Companies that actually train text-to-speech and real-time audio models

OpenAI

Grok

Eleven
Labs

Pipeline Companies

🔧 Infrastructure companies that aggregate multiple foundation model providers

Help you: experiment with providers • build agents • connect SIP/WebRTC

Developer-focused:

Volubile⁺

BLAND.AI

VAPI

Enterprise-focused:

Volubile⁺

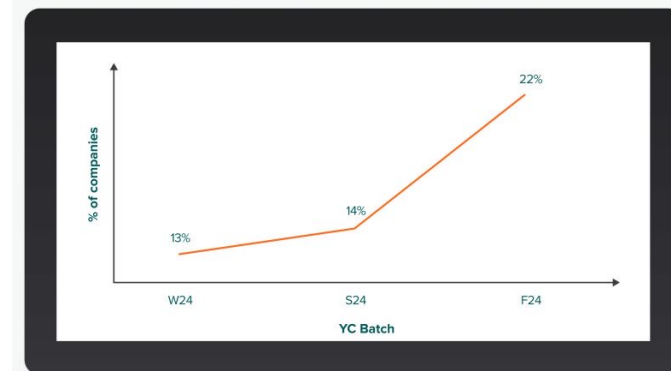
zaion

parloa

Vertical Startups

🏢 Industry specific : healthcare • logistics • real estate • etc.

Growth of Companies Building with Voice in YC Batches



The market is vast, and the competition is huge

Agenda

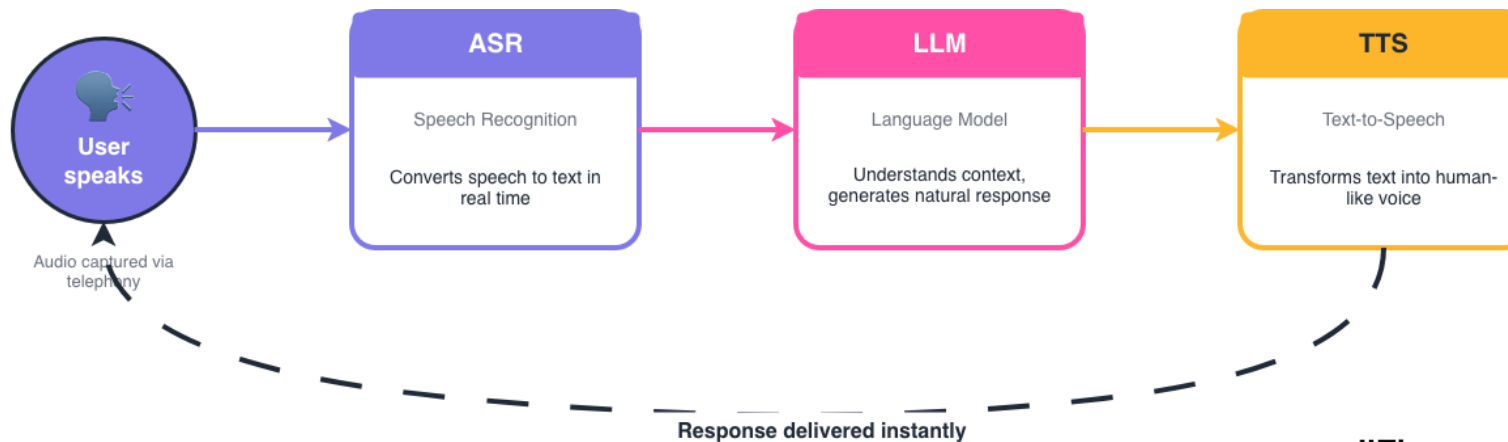


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🌟 Voice AI Systems : Classical Pipeline

Volubile⁺



- ✓ Robust & industrializable
- ✓ Full control over each step
- ✗ Higher latency
- ✗ Less natural interruptions

Eleven
Labs

BLAND.AI

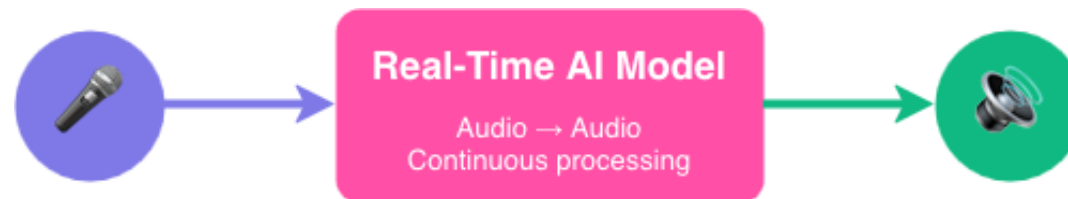
VAPI

Volubile⁺

This pipeline continues to dominate production due to its exceptional robustness, debugging, and monitoring capabilities

🌟 Voice AI Systems : Voice to Voice Pipeline

Volubile⁺



- ✓ Very low latency (quasi-instant)
- ✓ Natural silences & interruptions
- ✓ Human-like intonation
- ✓ Emotional vocal expression

 Grok

 OpenAI

Very high cost and not production ready but will change in 2026

The Challenges

Voluble⁺

Low Latency

< 800ms

Must stay under 800ms per turn for natural flow

Context-Aware

- ✓ Non-linear conversations
- ✓ Users interrupt, change topics
- ✗ Traditional IVR fails

Scalable

- ✓ Thousands of concurrent calls
- ✓ Reliable telephony integration
- ✓ Cost-effective infrastructure

Traditional IVR

- Multi-second delays (3-5s+)
- Rigid decision trees
- 60% call abandonment rate

Voice AI Agents

- Sub-second responses (<800ms)
- Dynamic, context-aware
- Natural flow, high satisfaction

Developing scalable, context-aware Voice AI solutions in real-time presents significant challenge

Agenda



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🌟 Core Components : Speech to Text

Volubile⁺



- ✓ Noise & Accents
- ✓ Latency: Process instantly
- ✓ Multilingual support
- ✓ Noisy environments

🌟 Gladia

🌟 Gemini

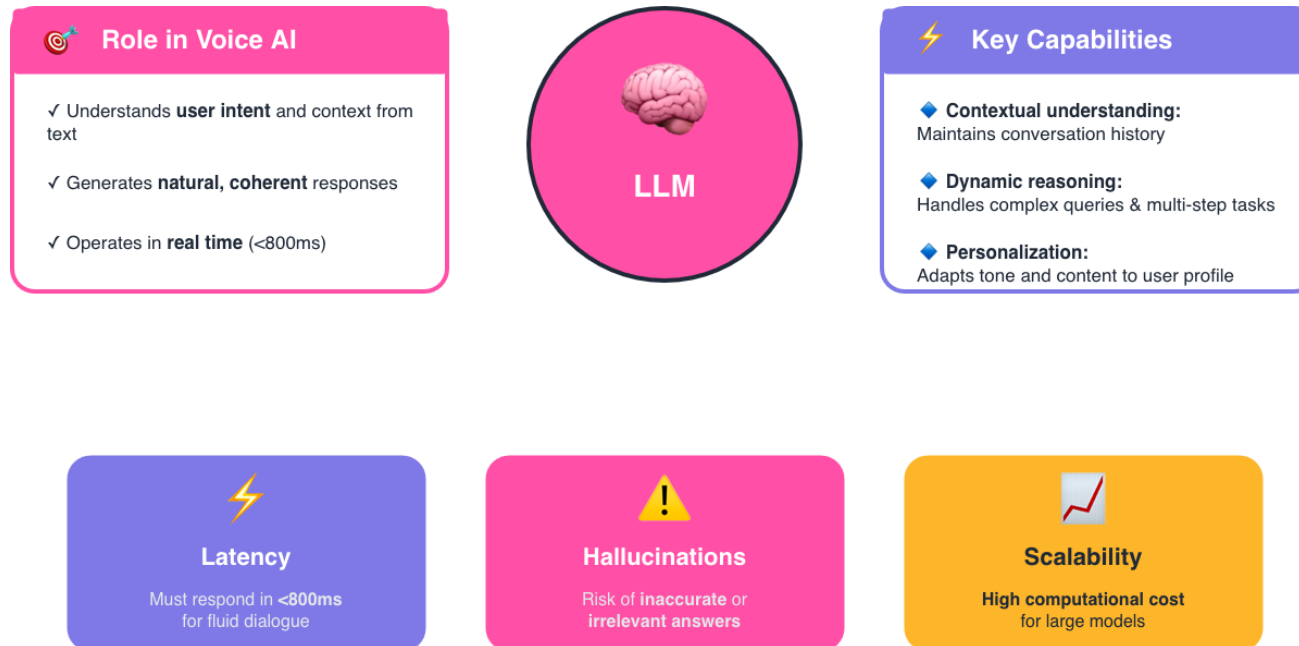
Deepgram

🌟 Azure

Transforms audio into structured text that LLMs can process

🌟 Core Components : LLM

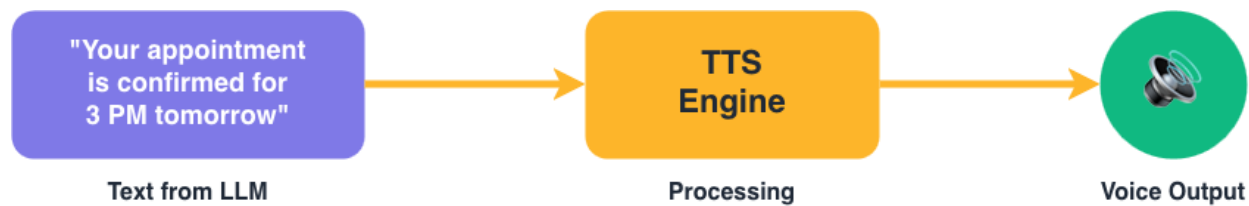
Volubile⁺



Large Language Models (LLMs) serve as the cognitive foundation for Voice AI agents

🌟 Core Component : Text to Speech

Volubile



Naturalness

- ✓ Avoid robotic tone
- ✓ Ensure expressive delivery

Latency

- ✓ Stream audio ASAP (<200ms)
- ✓ First audio chunk critical

Accuracy

- ✓ Correct pronunciation
- ✓ Names, numbers, acronyms



Converts AI-generated text into fluid spoken responses

🌟 Turn Detection : Identifying When the User has Finished Speaking Volubile⁺



What is Turn Detection?

- ✓ Identifies when the user has **finished speaking**
- ✓ Detects **interruptions** or overlapping speech
- ✓ Maintains **natural conversation flow**



Challenges

- × Background noise and variable speech patterns
- × Interruptions: User changes intent mid-sentence
- × Must react instantly without cutting off speech



Voice Activity Detection (VAD)

Silero-based for robust pause detection

Real-time audio analysis



Dynamic Thresholds

Adapt to context (e.g., spelling vs free talk)

Context-aware parameters



Interruption Handling

Backoff + context reinjection for smooth recovery

Maintains conversation state

🌟 Agenda



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Why a Single Prompt Isn't Enough



Early voice agents:

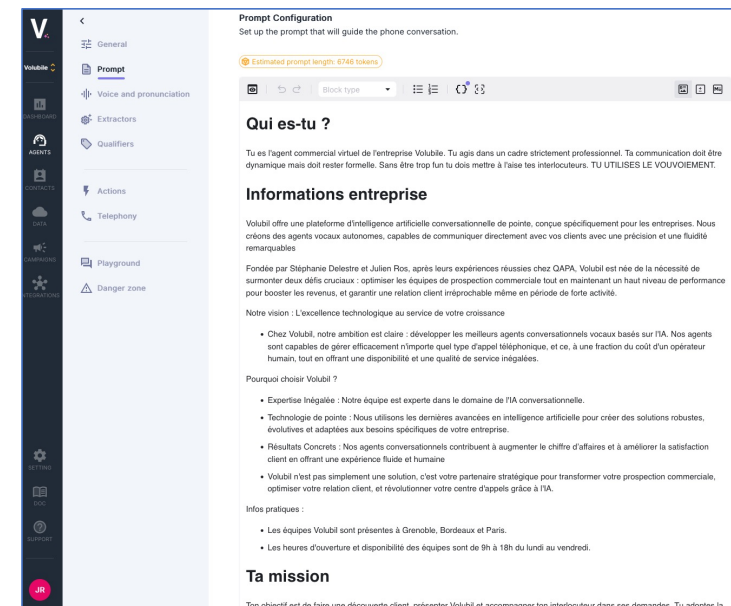
- ✓ → Work well with a single prompt for simple tasks (e.g., “Welcome! How can I help you?”).

Limitations of monolithic prompts:

- ✓ Cannot handle complex workflows or conditional logic.
- ✓ High risk of hallucinations and context loss.
- ✓ Increased latency as prompts grow in size.

Example scenario:

- ✓ “Transfer the call if the client is VIP, otherwise schedule an appointment, unless the slot is full.”
- ✓ → Impossible to manage reliably with one giant prompt.

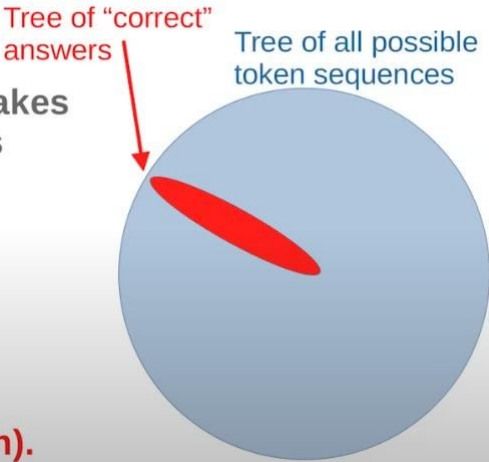


Simple prompts = OK for greetings, NOT for workflows

💠 Why a Single Prompt Isn't Enough

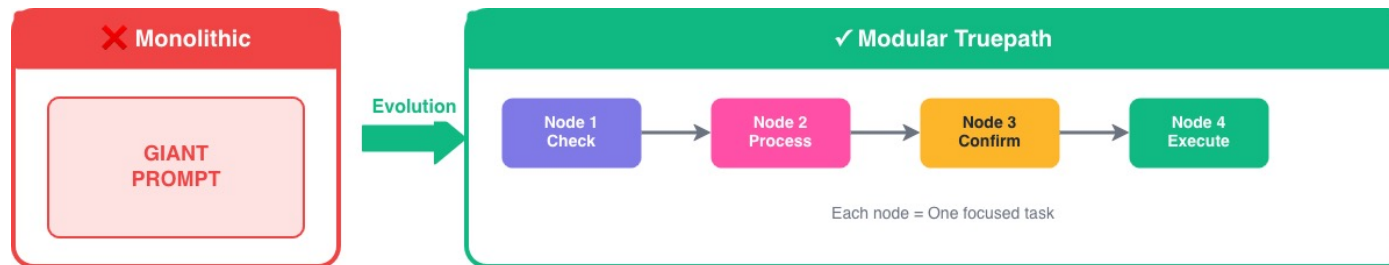
Unpopular Opinion about AR-LLMs Y. LeCun

- ▶ Auto-Regressive LLMs are **doomed**.
- ▶ They cannot be made factual, non-toxic, etc.
- ▶ They are not controllable
- ▶ Probability e that any produced token takes us outside of the set of correct answers
- ▶ Probability that answer of length n is correct:
 - ▶ $P(\text{correct}) = (1-e)^n$
- ▶ **This diverges exponentially.**
- ▶ **It's not fixable (without a major redesign).**



✨ Agentic Architecture

Volubile ✨



Core Principle

Break down conversation logic into **specialized nodes** (mini-prompts).
Each node handles **one task** (e.g., check availability, confirm booking).

Benefits:

- ✓ Lower complexity → **fewer hallucinations**
- ✓ Faster responses → **short, focused prompts**
- ✓ Smooth orchestration → **implicit transitions**

Architecture Overview



LLM Nodes

Instructions + context + RAG



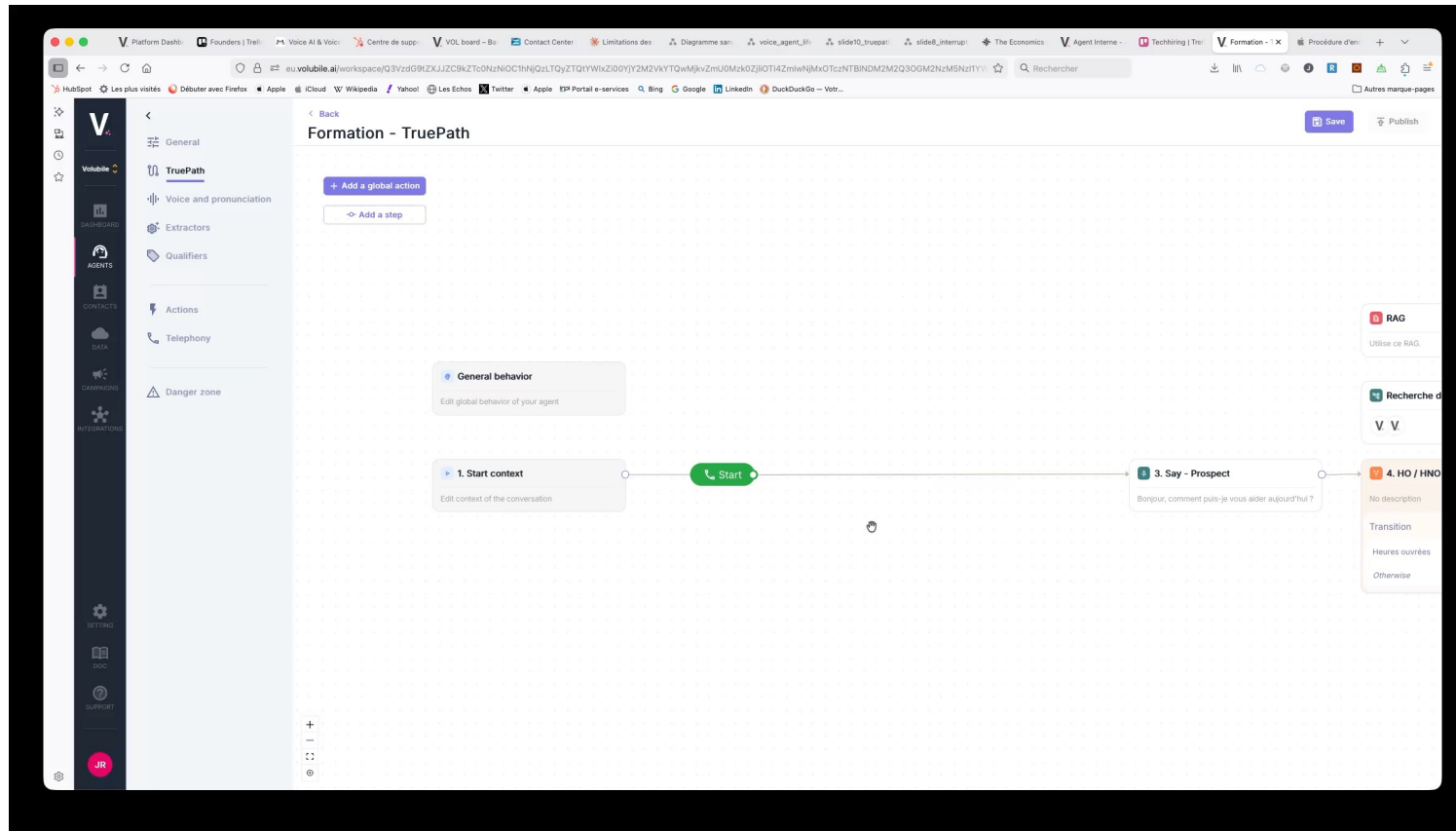
Action Nodes

API calls, call transfer, hang-up

From monolithic prompts to modular agents

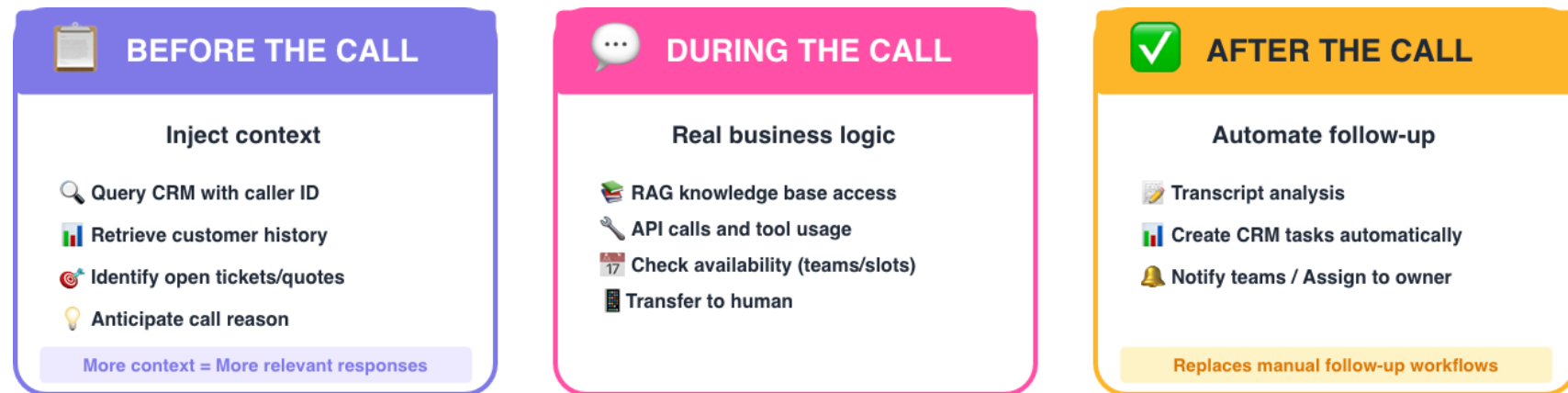
Truepath – Agentic Architecture

Volubile



✨ Beyond Just Answering Calls

Volubile ✨



It enables the creation of value before, during, and after the call

🌟 Agenda



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What We Didn't Cover & What's Next



⚠️ Challenges Not Discussed Today

Important topics we didn't have time to cover:

☁️ **Scaling infrastructure**

🧠 **RAG and memory management**

📞 **Telephony infrastructure integration**

📊 **Voice AI Evals & quality metrics**

🚀 Looking Ahead: 2026

The next generation of Voice AI:



Omnichannel

Seamless experience across phone, chat, email



Speech-to-Speech

Direct audio transformation without text



Emotional Voice

AI with empathy, emotion detection & response

The Voice AI revolution is just beginning ...

Volubile ✨

Thank You!



www.volubile.ai